

ALPER KALLE

Ph.D. Student in Computer Science

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Research Interests: Efficient Deep Learning · LLM Compression · Tensor Decomposition and Quantization · Edge AI

Education

- 2024 – Present **Ph.D. in Computer Science**, CEA (French Alternative Energies and Atomic Energy Commission) - Paris-Saclay University
Thesis: *Tensor Methods for Deep Neural Network Compression*
- 2021 – 2023 **Master in Applied Mathematics**, Toulouse III - Paul Sabatier University
- 2017 – 2021 **Bachelor in Mathematics**, Galatasaray University

Publications

- 2026 M.R.M. Boudjemaa, **A. Kalle**, X. Mai, J.H.M. Goulart, C. Févotte. "Whitening Spherical Gaussian Mixtures in the Large-Dimensional Regime". *IEEE International Conference on Acoustics, Speech, and Signal Processing* (Oral Presentation). **[Paper]**
- 2025 **A. Kalle**, T. Rudkiewicz, M.O. Ouerfelli, M. Tamaazousti. "Distribution-Aware Tensor Decomposition for Compression of Convolutional Neural Networks". *Advances in Neural Information Processing Systems (NeurIPS)*. **[Code][Paper]**

Research Experience

- Mar – Sep 2023 **Research Intern (Master)**, IRIT & IMT, Toulouse
Topic: *Analysis of tensor decomposition methods for machine learning in the large-data regime.*
○ Advisors: Henrique Goulart (IRIT) and Xiaoyi Mai (IMT).
- Mar – May 2025 **Teaching Assistant**, Polytech Paris-Saclay, Paris
Teaching assistant for the Mathematics 2 course; managed computer labs and student projects.

Summer Schools & Training

- Jun 2025 **A Journey through Deep Learning**, MaLGa Center, University of Genoa, Italy
40-hour intensive PhD-level school covering fundamentals of deep learning, CNNs, transformers, and generative models.

Skills

- Deep Learning PyTorch, TensorFlow, CUDA/cuDNN, Model Optimization
- Hardware GPU Programming, ONNX, TensorRT, Edge Deployment
- Programming Python, C#, MATLAB, SQL, Git, Slurm
- Languages Turkish (Native), English (Fluent), French (Fluent)

Selected Talks

- Oct 2025 **Seminar**, CEA, Paris
Title: *Compression of Neural Networks by Tensor Decomposition*
- Sept 2025 **Speech**, Junior Conference on Data Sciences and Engineering, Paris
Title: *Data-Driven Tensor Decomposition: A New Approach to Compressing CNNs*
- Jun 2023 **Seminar**, IRIT, Toulouse
Title: *Analysis of Tensor-Based Gaussian Mixture Model Estimation*

Awards & Scholarships

- 2021 – 2022 **Scholarship of Master Galatasaray**, Embassy of France in Turkey
- 2018 – 2021 **TUBITAK-2205 Excellence Scholarship**, TUBITAK (Turkey)
Merit scholarship for top undergraduate students in exact sciences.